

Comparative Transcriptomic Analysis of Skeletal Muscle Aging in Mice

A Multi-Muscle RNA-seq Case Study Using DESeq2, Biomarker Discovery and Functional Enrichment Analysis

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BACKGROUND

Sarcopenia is an age-associated decline in skeletal muscle mass and function. Different muscle types exhibit distinct susceptibility to aging; however, the molecular basis of these differences remains incompletely understood. This study compares transcriptomic changes across three skeletal muscle types to identify aging-associated genes and pathways.

DATASET

GEO Accession: GSE287832
Organism: Mus musculus
Samples: 18 RNA-seq samples
Muscles:

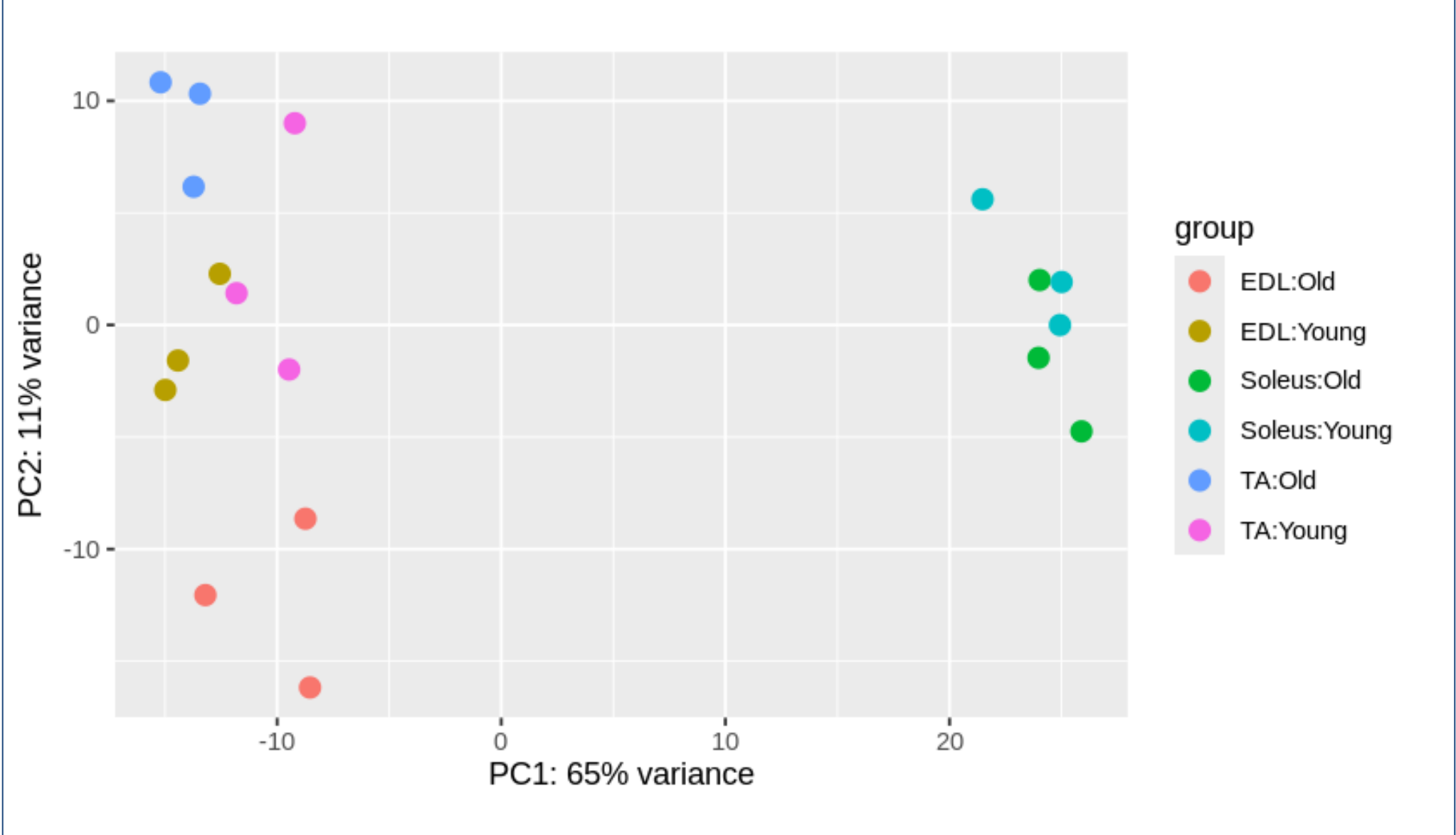
- EDL
- Soleus
- TA

Conditions:

- Young
- Old

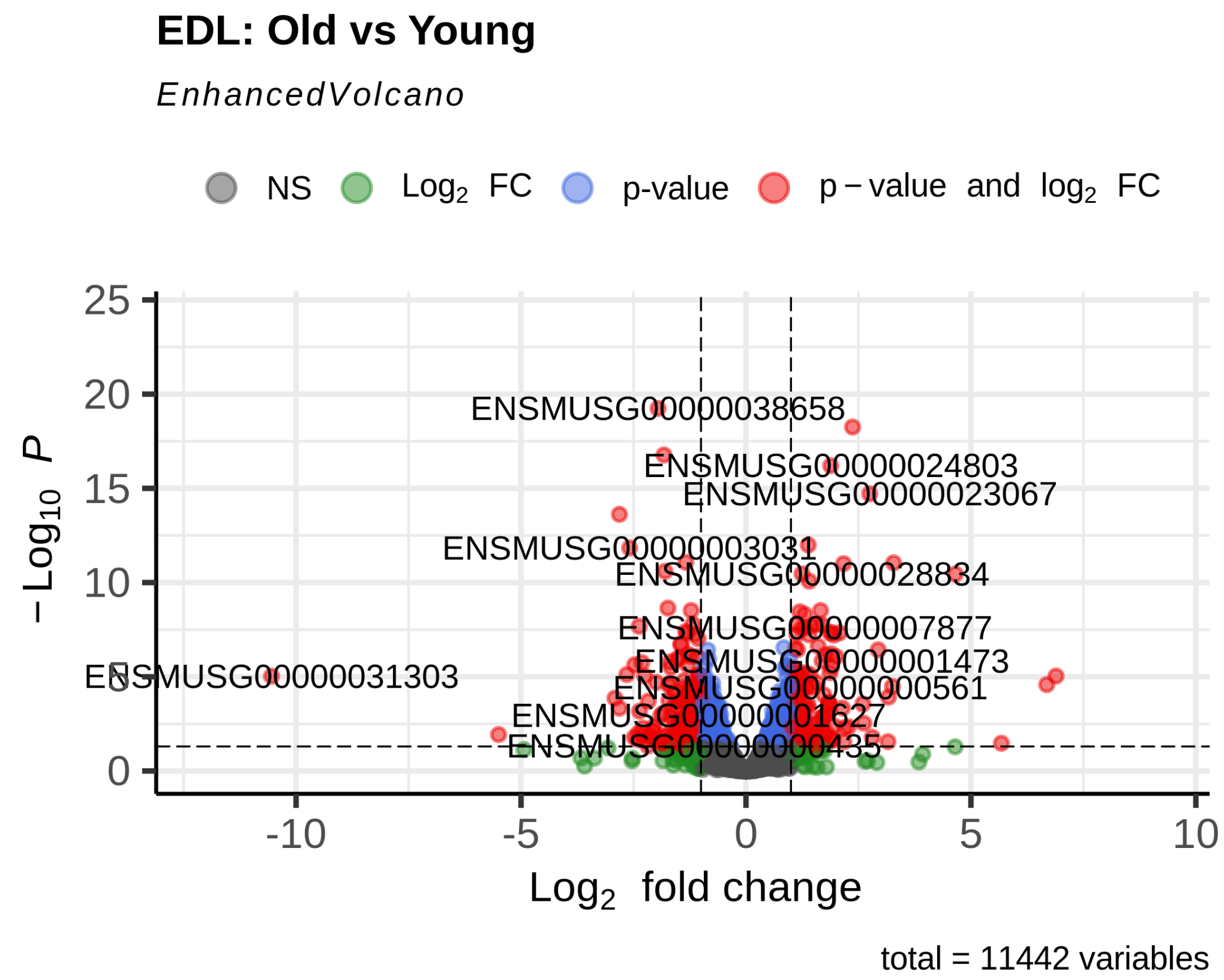
Replicates: 3 biological replicates per group

PCA Analysis



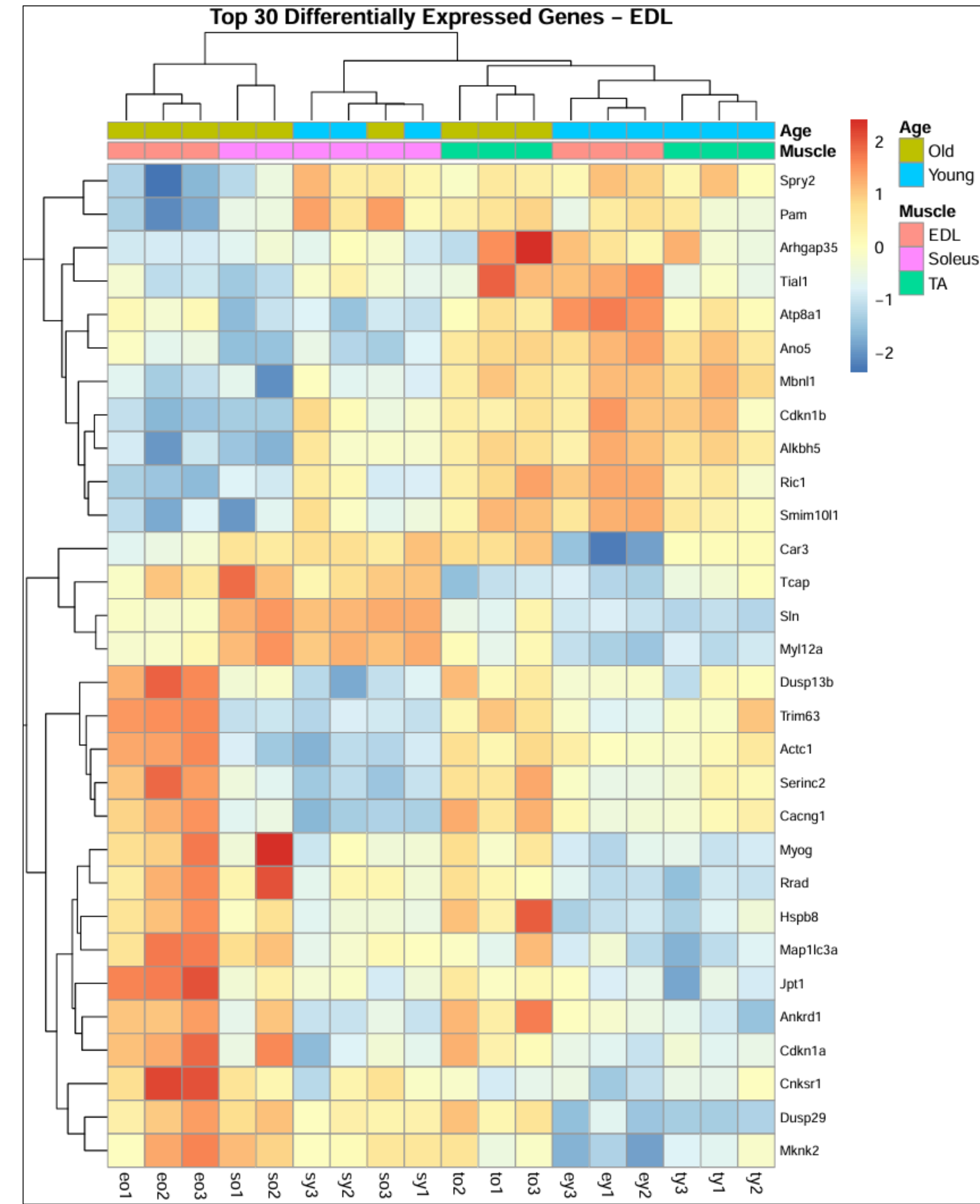
Samples cluster according to muscle type and age, indicating robust biological separation.

EDL Volcano Plot



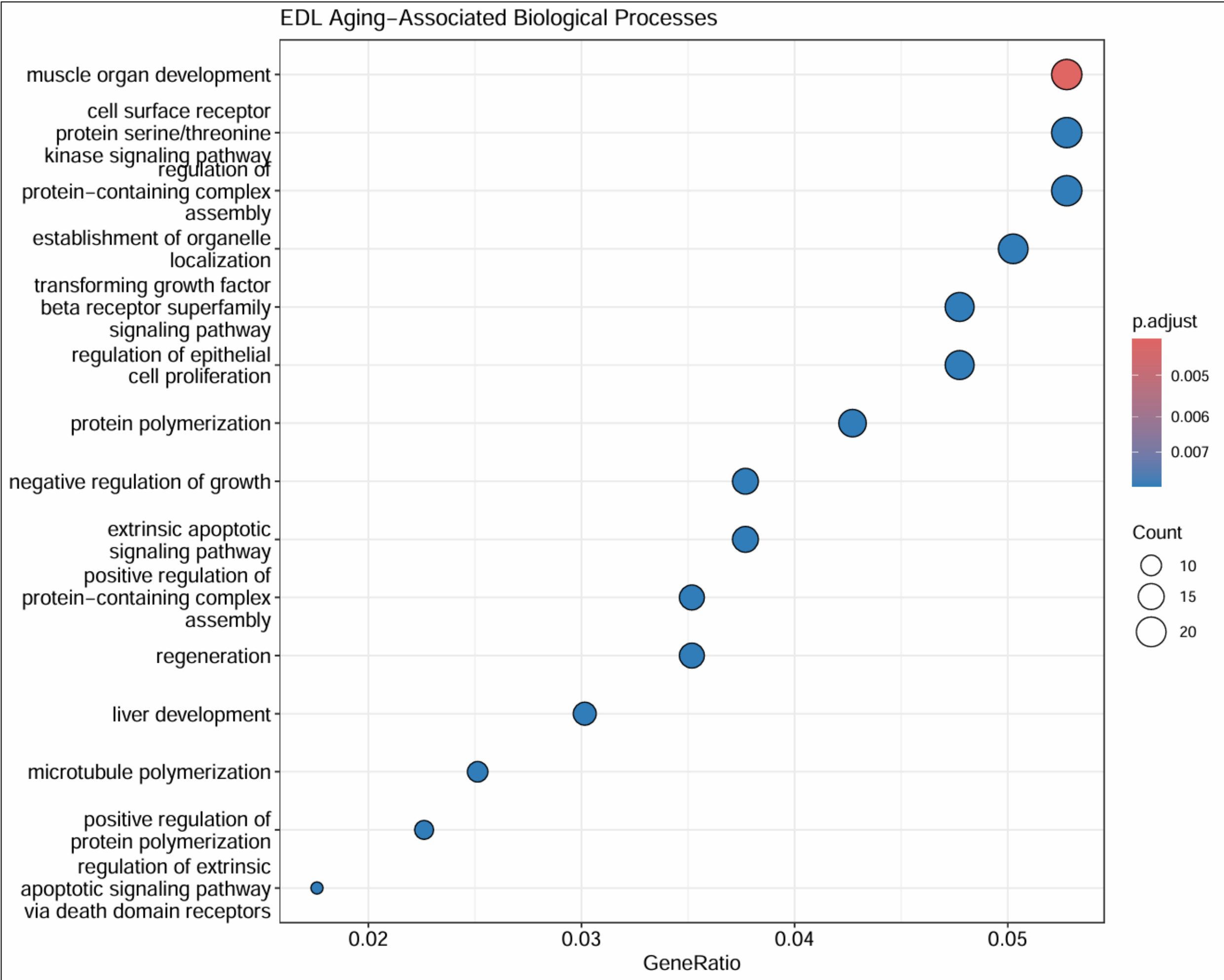
Aging induces extensive transcriptional remodeling in EDL muscle.

Top EDL Biomarkers



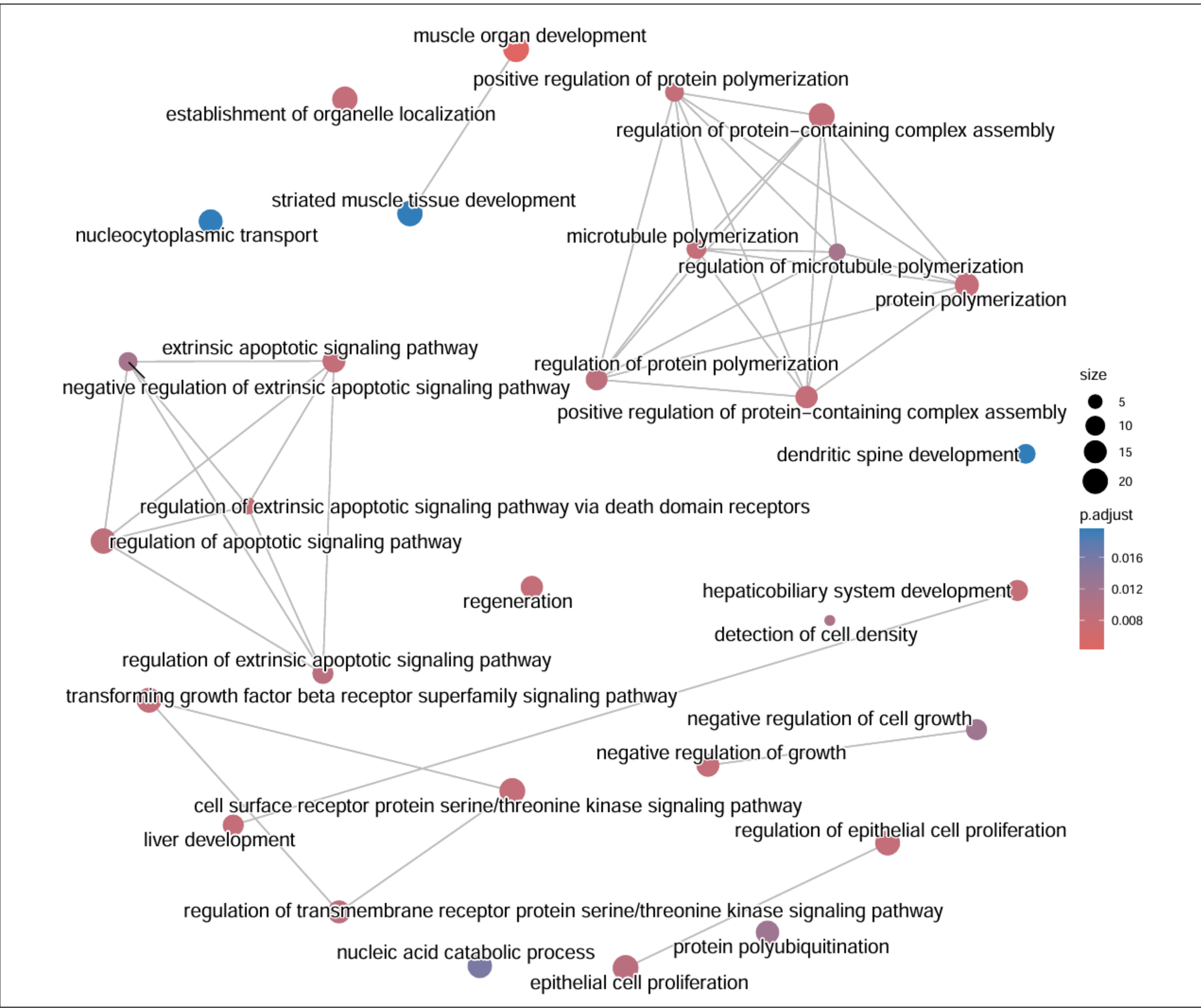
Top biomarkers clearly distinguish young and aged EDL samples.

EDL GO Enrichment

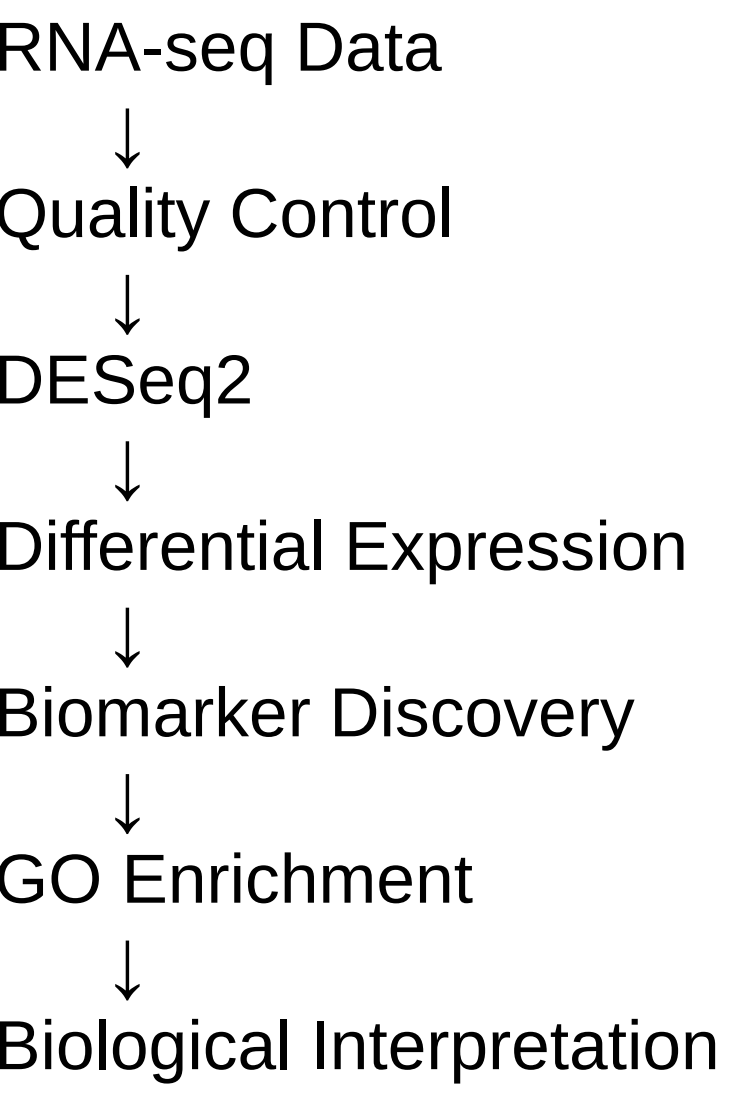


Functional enrichment reveals aging-associated pathways related to muscle development, growth regulation and cellular remodeling.

EDL Pathway Network

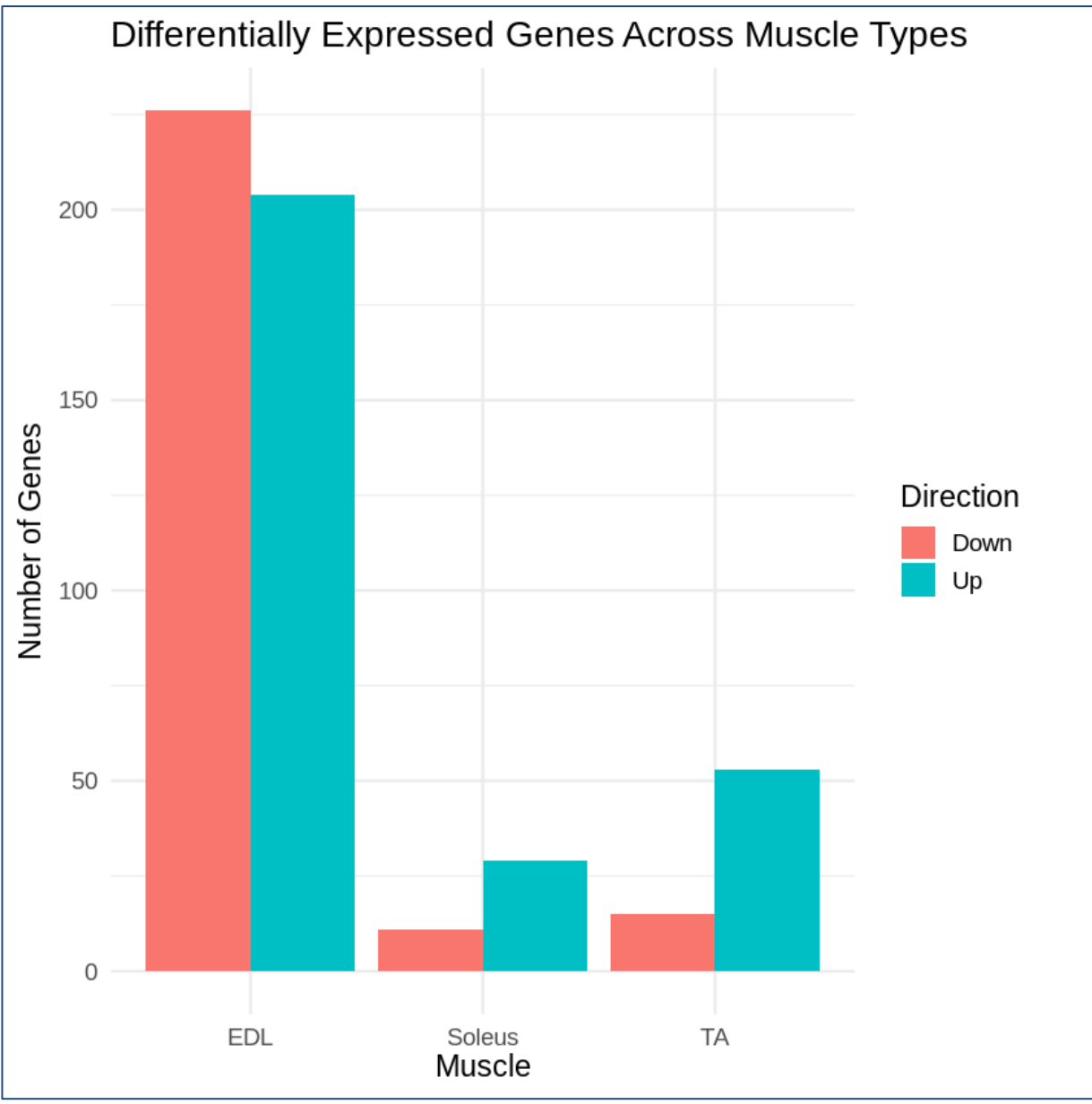
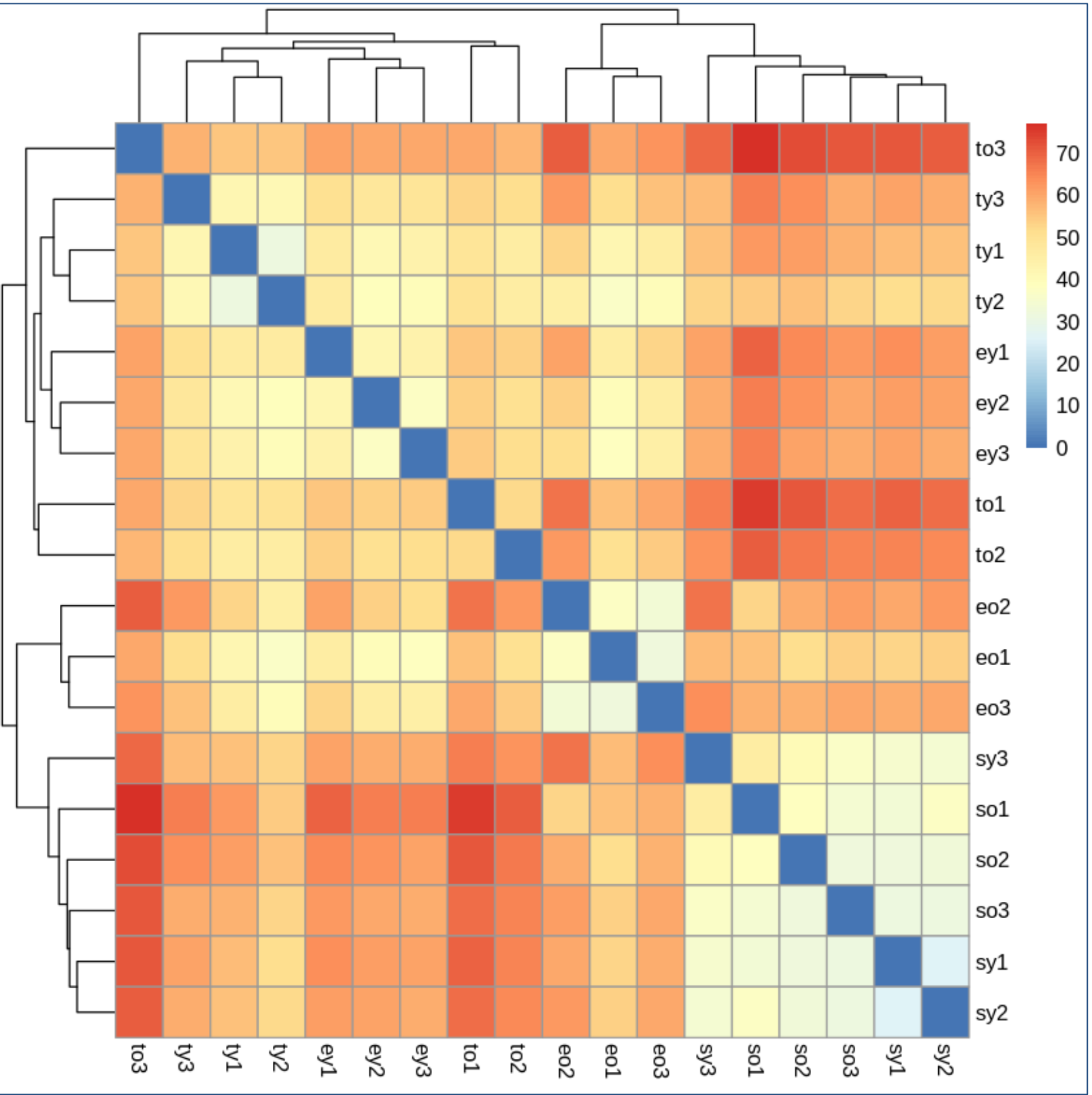


Pathway network analysis highlights interconnected biological processes altered during aging.



Sample Distance Heatmap

Biological replicates exhibit strong within-group similarity and no major outliers.



Differential Expression Summary

EDL exhibited the strongest aging-associated transcriptional response.

KEY FINDINGS

- EDL displayed 1129 significant genes.
- Soleus displayed only 44 significant genes.
- TA displayed 105 significant genes.
- EDL showed strong enrichment of growth regulation, apoptosis and developmental pathways.
- Soleus exhibited vascular and extracellular matrix adaptation pathways.
- TA maintained muscle-specific adaptive programs.
- Aging affects skeletal muscles in a muscle-type-specific manner.

CONCLUSIONS

- Aging responses differ substantially across muscle types.
- EDL is highly susceptible to transcriptomic remodeling.
- Soleus exhibits relative resilience.
- TA demonstrates an intermediate phenotype.
- Distinct biological pathways underlie muscle-specific aging.

SKILLS DEMONSTRATED

- ✓ RNA-seq Analysis
- ✓ Differential Expression (DESeq2)
- ✓ Biomarker Discovery
- ✓ Data Visualization
- ✓ Functional Enrichment
- ✓ Scientific Reporting
- ✓ Biological Interpretation
- ✓ Research Communication